

Best practices in computing

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Applied Quantitative Methods II
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Today's goals

- Understand why computing practices matter for research

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- Write better R code: functions, checks, and style
- Understand the role of plain text and command line tools
- Deepen your knowledge of version control with Git

Roadmap

Introduction

Project Organization

Writing Better Code

Plain Text and Tools

Version Control with Git

Wrap-up

Code is a scientific artifact

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- The standards of rigor you apply to your theory and research design should extend to your computing

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 - Running scripts in the wrong order
- **Efficiency:** time invested in workflow saves time later

You come back to a project after 6 months.

You need to update one figure.

How long does it take you?

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- Never overwrite it, never modify it by hand
- Code transforms raw data into everything else

Organizing a project: one example

```
PROJECT_FOLDER/  
├── analyses/  
│   ├── analyze.R  
│   └── output/  
│       └── table_models.tex  
├── create_data/  
│   ├── data.R  
│   └── output/  
│       └── data.csv  
├── document/  
│   ├── doc.tex  
│   ├── doc.pdf  
│   ├── doc.aux  
│   └── ...  
├── plots/  
│   ├── output/  
│   │   └── scatter.pdf  
│   └── plots.R  
├── Makefile  
└── README.md
```

- Full example: github.com/franvillamil/workflow_example

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- Delete all output, re-run, and get the same results

File naming conventions

Bad	Good
Final Data.csv	data_cleaned.csv
Datos educación.csv	datos_educacion.csv
analysis.R (which one?)	01_clean_data.R
figure1final2.pdf	fig_scatter_income.pdf
My Thesis Draft (3).docx	thesis_draft.tex

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- Use **descriptive names**: what is in the file, not when you made it

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 - `doc.tex`: `\input{../analyses/output/table_models}`

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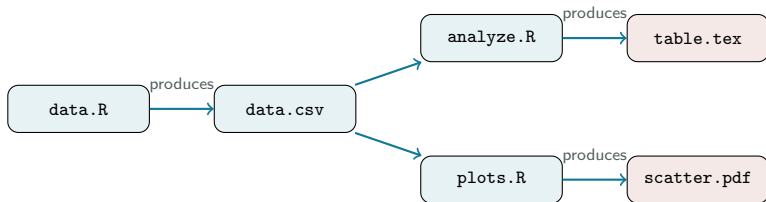
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- No local $\text{\LaTeX}$? Use **Overleaf** — same `\input{}` workflow

Automating the pipeline: Makefiles

```
all:  create_data/output/data.csv analyses/output/table_models.tex \
      plots/output/scatter.pdf
create_data/output/data.csv:  create_data/data.R
                              Rscript --no-save create_data/data.R
analyses/output/table_models.tex:  analyses/analyze.R \
                                   create_data/output/data.csv
                              Rscript --no-save analyses/analyze.R
plots/output/scatter.pdf:  plots/plots.R \
                           create_data/output/data.csv
                              Rscript --no-save plots/plots.R
```

Note: Makefiles require **tabs** (not spaces) for indentation

Makefiles: the logic



- make only re-runs what has **changed**
- The dependency graph ensures correct ordering

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 - Producing plots with consistent formatting

Define constants at the top

Bad	Good
<pre>df = df[df\$year >= 2000,] ... df2 = df2[df2\$year >= 2000,]</pre>	<pre>start_year = 2000 ... df = df[df\$year >= start_year,] df2 = df2[df2\$year >= start_year,]</pre>

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- Change once, applies everywhere
- From the example project:
 - `n_obs = 1000`
 - Used throughout — change it once to re-run with different sample size

Write checks and assertions

```
# After merging two datasets
merged = merge(df1, df2, by = "id")
if(nrow(merged) != nrow(df1)) {
  stop("Merge changed number of rows!")
}

# Check for duplicates
if(any(duplicated(df$id))) {
  stop("Duplicate IDs found!")
}

# Sanity check on values
if(any(df$age < 0 | df$age > 120, na.rm = TRUE)) {
  warning("Suspicious age values detected")
}
```

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- This is especially valuable when running scripts via `Rscript` or `make`

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- Not paranoia — this has caused real problems in published work

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- Write **comments** for non-obvious code
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 - Use section dividers: `# =====`
- Your code is read more often than it is written

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- “It works on my machine”
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- “I fixed a few values in Excel”
 - Undocumented, unreproducible, invisible to collaborators

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 - SPSS menus, Excel transformations, manual edits
- “It works on my machine”
 - Almost always an absolute path problem
- “I fixed a few values in Excel”
 - Undocumented, unreproducible, invisible to collaborators
- Scripts that must be run in a specific undocumented order

Roadmap

Introduction

Project Organization

Writing Better Code

Plain Text and Tools

Version Control with Git

Wrap-up

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- The goal: use it for **as much as possible**

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- Terminal on Mac/Linux, Git Bash or WSL on Windows

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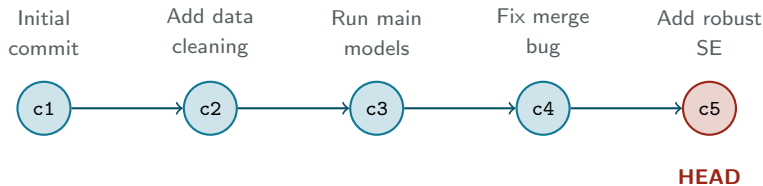
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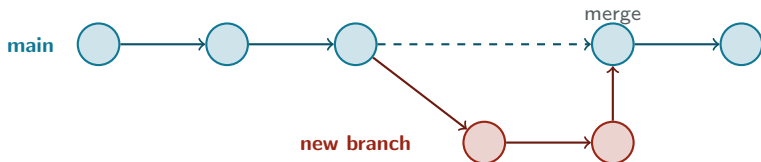
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- Today: going deeper into **good practices**

Git: commits as snapshots



- Each **commit** is a snapshot of your project at a point in time
- The **history** is a chain of commits, each with a message
- HEAD is a pointer to your current position

Branches: parallel lines of work



- A **branch** is an independent line of commits
- Useful for trying changes without breaking `main`
- **Merge** brings the work back together

The .gitignore file

Track (source files):

- R scripts (.R)
- L^AT_EX source (.tex)
- Makefiles
- Documentation (.md)
- Small data files (.csv)

Do NOT track:

- Generated output (.pdf, .png)
- L^AT_EX auxiliary files (.aux, .log)
- Large data files
- System files (.DS_Store)
- Sensitive data

```
.gitignore:
```

```
*.pdf
```

```
*.aux
```

```
*.log
```

```
.DS_Store
```

```
output/
```

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 - You should be able to reconstruct what you did and why

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- Example: github.com/franvillamil/vox_military

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- Even for solo work: syncing between laptop and desktop

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Key resources

- Kieran Healy, *The Plain Person's Guide to Plain Text Social Science*:
→ plain-text.co
- MIT, *The Missing Semester of Your CS Education*:
→ missing.csail.mit.edu
- Software Carpentry lessons (Unix Shell, Git):
→ software-carpentry.org/lessons
- Bruno Rodrigues, *Building Reproducible Analytical Pipelines with R*:
→ raps-with-r.dev

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- **Use plain text:** portable, versionable, scriptable
- **Use Git:** commit often, write good messages, share via GitHub

For next week

- Complete Assignment 10 (computing practices)
- Next session: In-class exam + course review
 - Exam covers sessions 1–9
 - Second half: review and open questions

Questions?